

Sugar-Dark Side of Sweetness (cont...)

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Rank the following (Good → bad)



1



3



2

Rank the following (Good → bad)



Liver detoxify substances that are harmful for your body



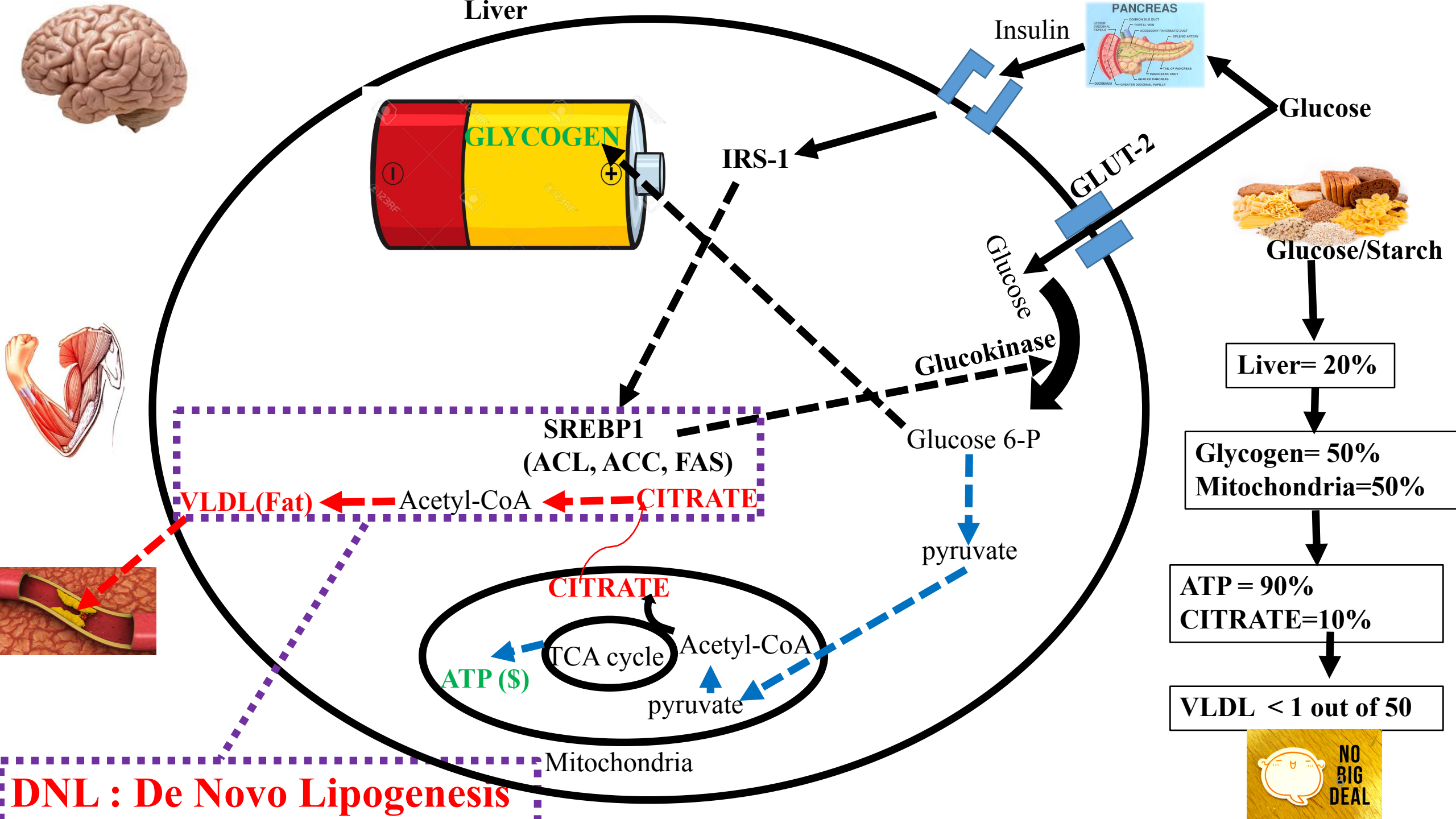
**How a non toxic carbohydrate is metabolized ?
What happens in the Liver ?**



Before I start.....



**You don't have to remember NAMES...
Just follow pathway (Flow of food through
the lever cells).**



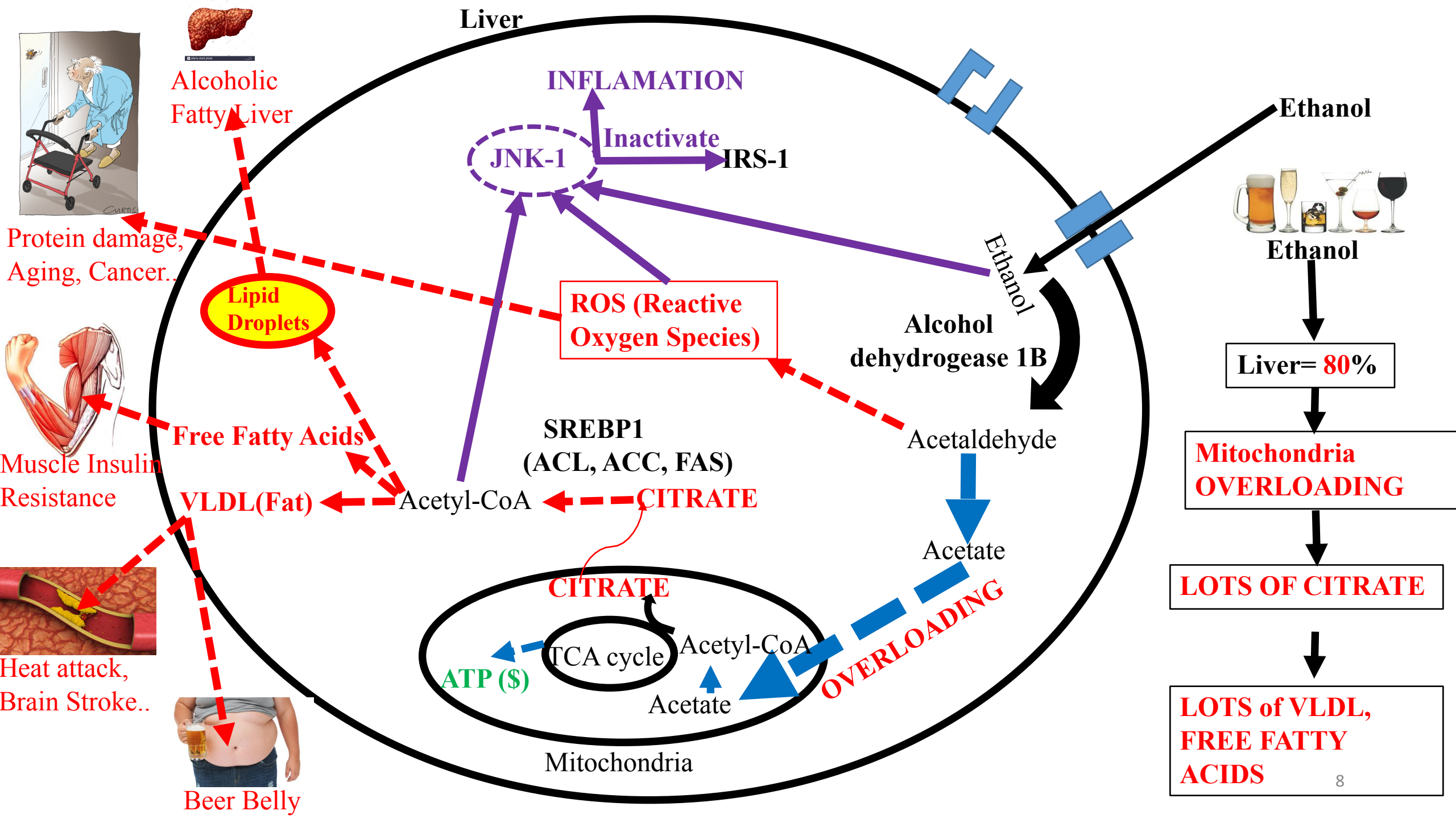
DNL : De Novo Lipogenesis

How alcohol is metabolized in the Liver ?



Liver detoxify substances that are harmful for your body

- ❖ Alcohol is not necessary for biological processes. Hence considered as toxin and mostly processed in the Liver.
- ❖ Alcohol processing (Kidney+muscle+brain= 10%, Intestine+stomach= 10%, LIVER = **80%**)
- ❖ **LIVER SHOULD WORK 4 TIMES MORE THAN THE CASE OF NON-TOXIC CARBOHYDRATE**
- ❖ **ETHANOL DOES NOT NEED INSULIN TO ENTER (Diffusion).**

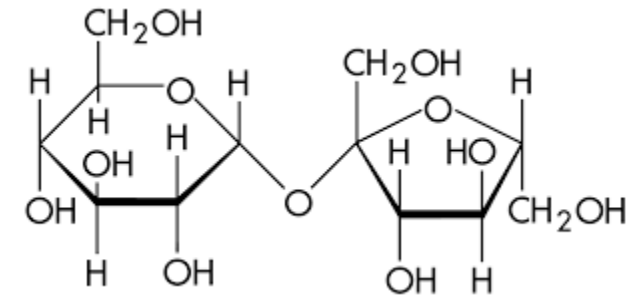


How Sugar is metabolized in the Liver ?

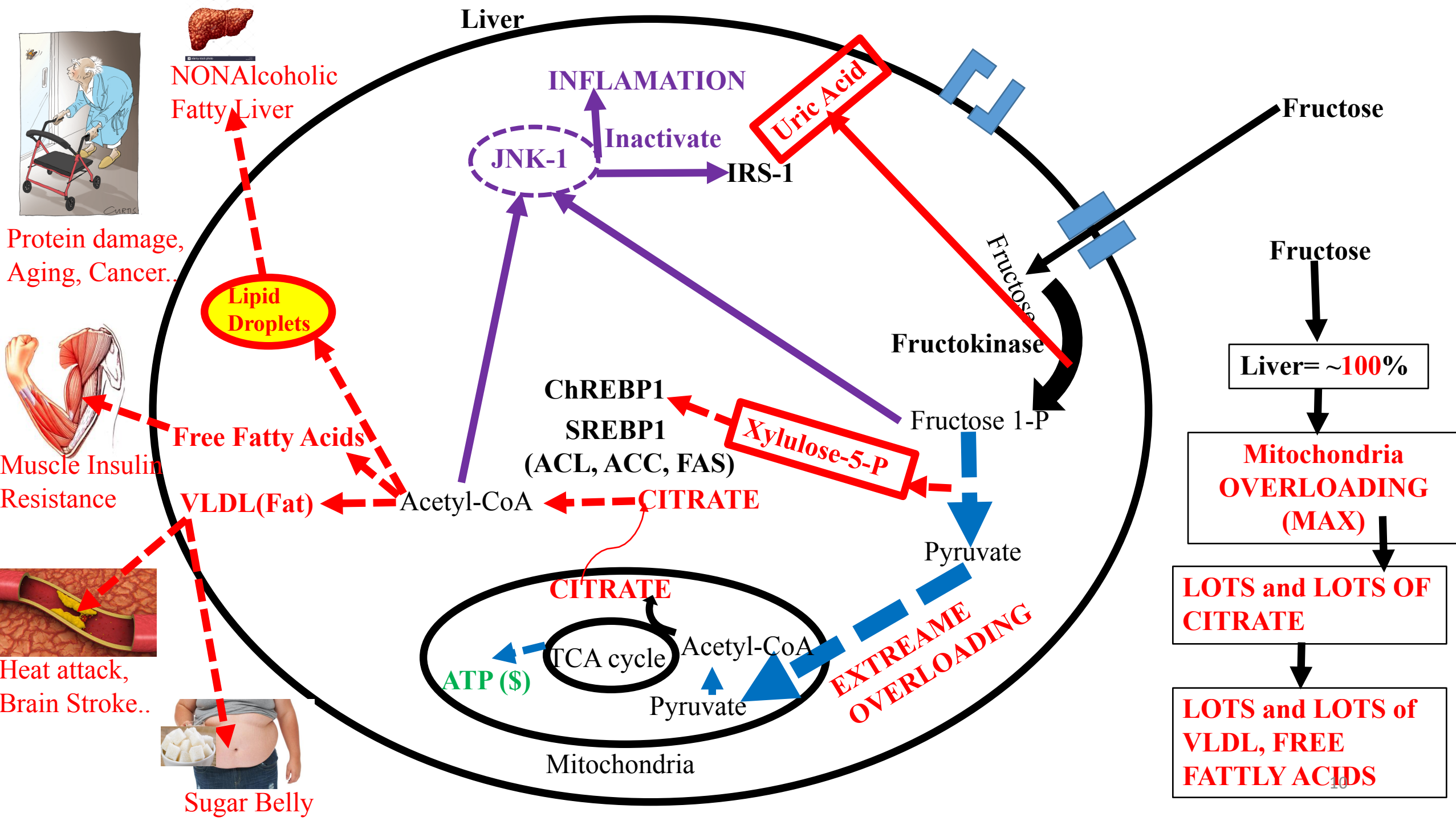


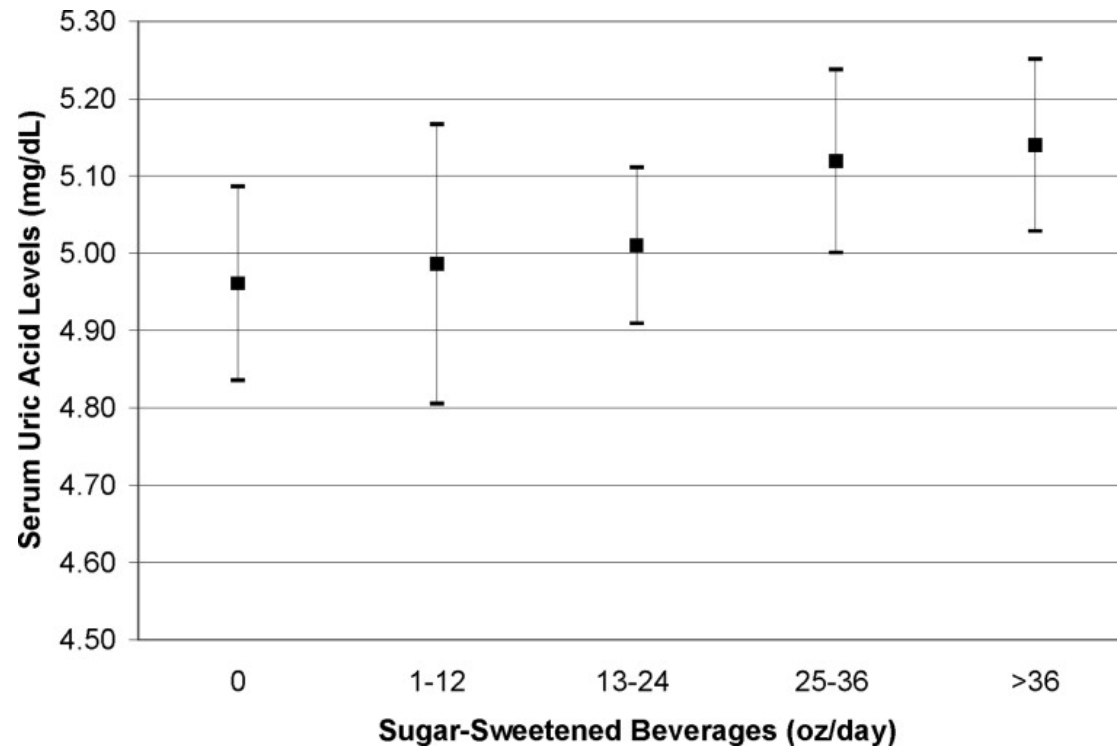
Liver detoxify substances that are harmful for your body

❖ **Sugar = 50%Glucose+50%Fructose (LECTURE 3).**



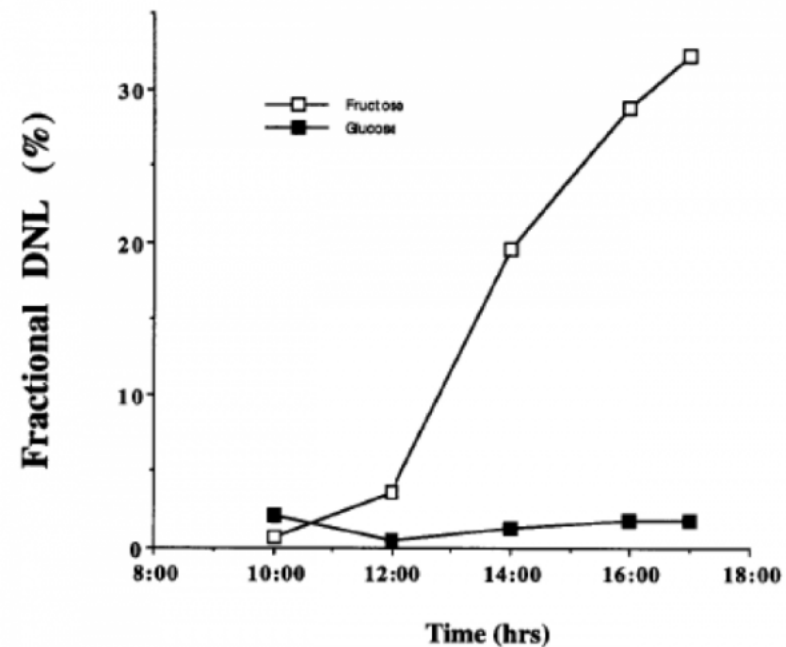
- ❖ ***Fructose is not necessary for the body → Toxin → ~100% Processed in the Liver.***
- ❖ **LIVER SHOULD WORK 5 TIMES MORE**
- ❖ **DOES NOT NEED INSULIN TO ENTER (Diffusion).**





Nguyen, Stephanie, et al. "Sugar-sweetened beverages, serum uric acid, and blood pressure in adolescents." *The Journal of pediatrics* 154.6 (2009): 807-813.

Conclusion: Fructose is linked to Uric acid, blood pressure



Hollorstein et al. *Ann Rev Nutr.*
16:523, 1996

The screenshot shows the Food Safety News (FSN) website. The main article is titled "The American Diet: A Sweet Way to Die" by Bruce Clark, dated February 26, 2010. The article text visible includes: "I am often asked what foods I avoid because they may carry a risk of contamination with pathogenic organisms. I have a very short list of foods to avoid on principle based on concerns about the potential for bacterial adulteration. But I am increasingly concerned about the general quality of the American diet as a cause of". To the right of the article is an advertisement for Eurofins, stating "A commitment to food safety starts with your employees." and "Find out more about our food safety training courses". The browser's address bar shows the URL: <https://www.foodsafetynews.com/2010/02/the-am>. The browser's search bar contains the text "es denovo lypogenesis".

Conclusion: Fructose increases de novo lipogenesis

BMJ Open Fructose intake and risk of gout and hyperuricemia: a systematic review and meta-analysis of prospective cohort studies

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Title: Jamnik J, Rehman S, Blanco Mejia S, et al. Fructose intake and risk of gout and hyperuricemia: a systematic review and meta-analysis of prospective cohort studies. *BMJ Open* 2016;6:e013191. doi:10.1136/bmjopen-2016-013191

► Prepublication history and additional material is available. To view please visit the journal (<http://dx.doi.org/10.1136/bmjopen-2016-013191>).

Received 9 July 2016
Revised 25 August 2016
Accepted 5 September 2016



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ABSTRACT

Background: The prevalence of hyperuricemia and gout has increased in recent decades. The role of dietary fructose in the development of these conditions remains unclear.

Objective: To conduct a systematic review and meta-analysis of prospective cohort studies investigating the association fructose consumption with incident gout and hyperuricemia.

Design: MEDLINE, EMBASE and the Cochrane Library were searched (through September 2015). We included prospective cohort studies that assessed fructose consumption and incident gout or hyperuricemia. 2 independent reviewers extracted relevant data and assessed study quality using the Newcastle-Ottawa Scale. We pooled natural-log transformed risk ratios (RRs) using the generic inverse variance method. Interstudy heterogeneity was assessed (Cochran Q statistic) and quantified (I^2 statistic). The overall quality of the evidence was assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach.

Results: 2 studies involving 125 299 participants and 1533 cases of incident gout assessed the association between fructose consumption and incident gout over an average of 17 years of follow-up. No eligible studies assessed incident hyperuricemia as an outcome. Fructose consumption was associated with an increase in the risk of gout (RR=1.62, 95% CI 1.28 to 2.03, $p<0.0001$) with no evidence of interstudy heterogeneity ($I^2=0\%$, $p=0.33$) when comparing the highest (>11.8% to >11.9% total energy) and lowest (<6.9% to <7.5% total energy) quantiles of consumption.

Limitations: Despite a dose-response gradient, the overall quality of evidence as assessed by GRADE was low, due to indirectness. There were only two prospective cohort studies involving predominantly white health professionals that assessed incident gout, and none assessed hyperuricemia.

Conclusions: Fructose consumption was associated with an increased risk of developing gout in predominantly white health professionals. More prospective studies are necessary to understand

Strengths and limitations of this study

- This systematic review and meta-analysis assessed the overall quality of the evidence using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach.
- Large prospective cohort studies that were of high quality and had a long duration of follow-up were included.
- The pooled results showed good consistency (low between-study heterogeneity) and evidence of a dose-response gradient.
- Only two prospective cohort studies with low external generalisability were available for inclusion.
- The observational design of the prospective cohort studies did not allow for causal inferences to be drawn.

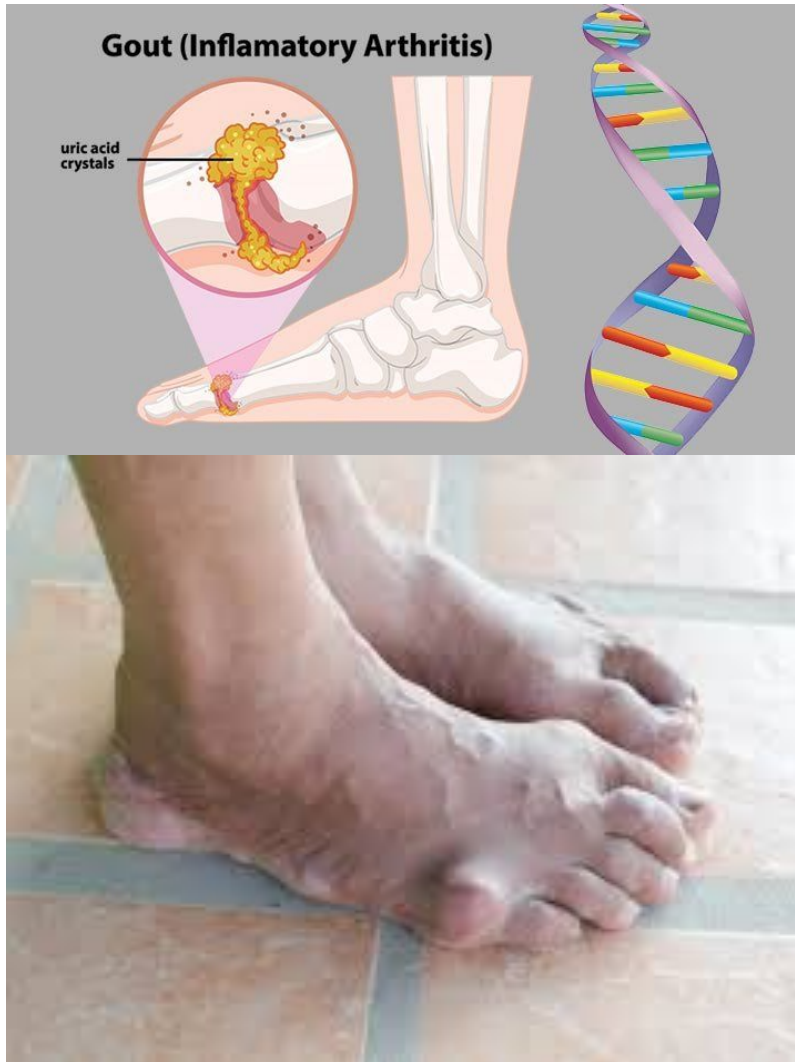
better the role of fructose and its food sources in the development of gout and hyperuricemia.

Protocol registration number: NCT01608620.

INTRODUCTION

Gout is a systemic rheumatic condition characterised by monosodium urate crystal deposition and accumulation around joints. Individuals with gout often experience acute and recurring attacks of arthritis that can affect several joints.¹ Hyperuricemia or excessive circulating concentrations of urate, the final product of purine metabolism, is a major risk factor for gout and plays a major role in the pathogenesis of this condition.² Chronic hyperuricemia and gout also represent potential risk factors for cardiovascular disease (CVD).³ According to the National Health and Nutritional Examination Survey 2007–2008, hyperuricemia affects >20% of

Conclusion



Saben, Jessica L., et al. "Excess maternal fructose consumption increases fetal loss and impairs endometrial decidualization in mice." *Endocrinology* 157.2 (2015): 956-968.

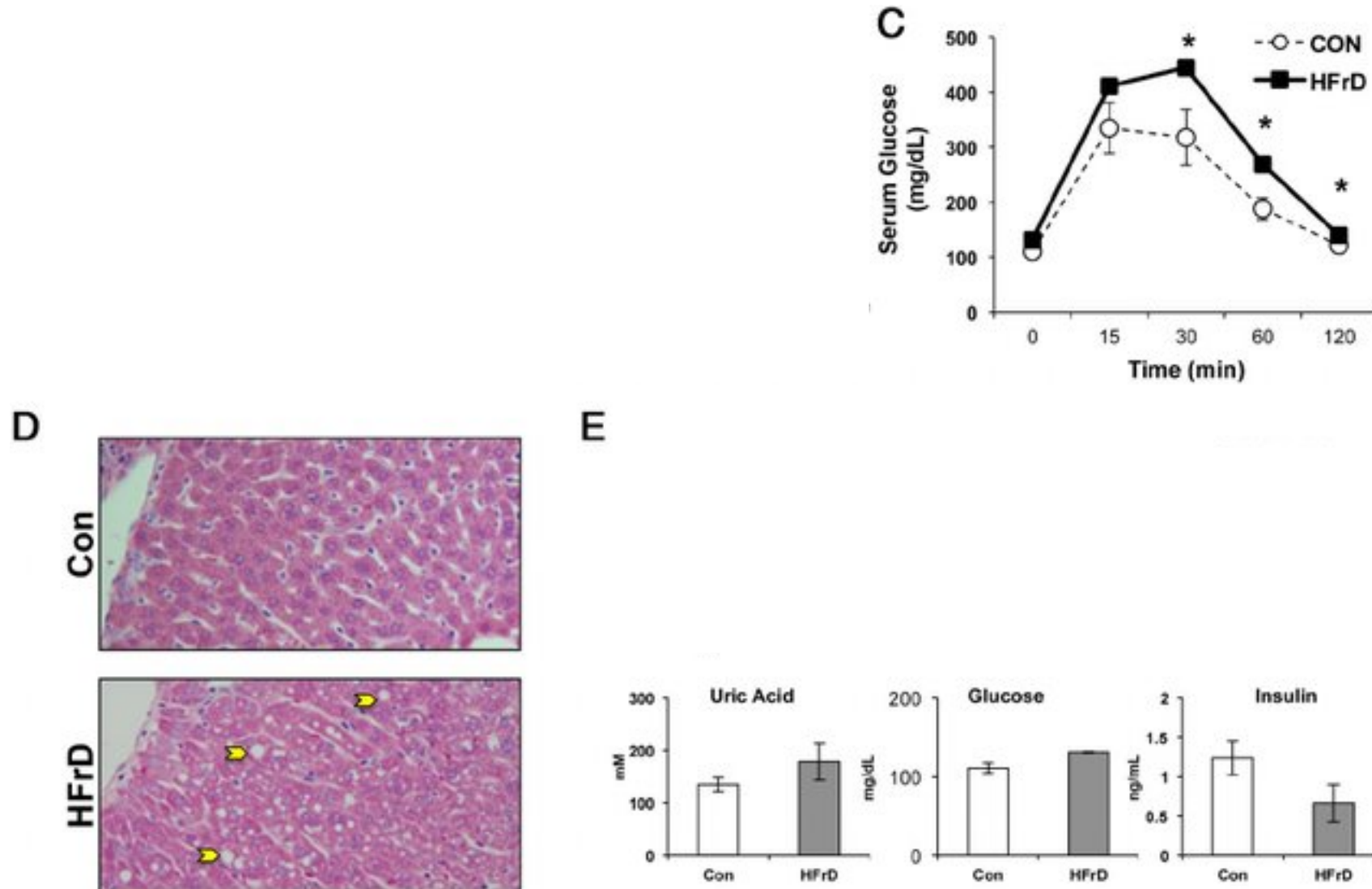


Figure 1. **Six weeks of high fructose consumption** induces fatty liver and glucose intolerance. Six-week-old C57BL/6 mice were fed either control (Con) or 60% fructose diet (HFrD) for six weeks, and then their metabolic profile was characterized. C : Glucose tolerance test performed over 120 minutes.

D: H&E staining of livers from Con- and HFrD-fed mice. Yellow arrows indicate lipid droplets.

E: Serum measures in fasted mice: uric acid, glucose, and insulin.

Conclusion: Fructose(↑) = FAT(↑), Blood glucose (↑), Uric Acid(↑)

Drug to treat Gout

Effect of Allopurinol on Blood Pressure of Adolescents With Newly Diagnosed Essential Hypertension

A Randomized Trial

Daniel I. Feig, MD, PhD

Beth Soletsky, RN

Richard J. Johnson, MD

HYPERTENSION IS COMMONLY associated with hyperuricemia.^{1,2} Early investigators proposed uric acid as having a causal role in hypertension.³⁻⁵ However, an elevation of uric acid in hypertension could be a consequence of reduced renal function, the use of diuretics, the presence of hyperinsulinemia and oxidative stress, or elevated renal vascular resistance, which are commonly present in this condition.⁶ As such, hyperuricemia is not considered a true risk factor for hypertension by the Joint National Committee,⁷ nor is it considered a cardiovascular risk factor by most expert organizations.⁸

Recent studies have challenged this long-standing paradigm. For example, numerous studies have reported that hyperuricemia independently predicts the development of hypertension,⁹⁻¹³ even in individuals lacking features of the metabolic syndrome.¹⁴ If hyperuricemia precedes the development of hypertension then it cannot simply be a secondary phenomenon. We also previously reported that elevated uric acid is present in nearly 90% of adolescents presenting with essential hypertension.¹⁵ Of 63 participants with essential hypertension, 89% had a uric acid level higher than 5.5 mg/dL (mean, 6.7 mg/dL; to convert milligrams per deciliter to micro-

Context Hyperuricemia is a predictor for the development of hypertension and is commonly present in new-onset essential hypertension. Experimentally increasing uric acid levels using a uricase inhibitor causes systemic hypertension in animal models.

Objective To determine whether lowering uric acid lowers blood pressure (BP) in hyperuricemic adolescents with newly diagnosed hypertension.

Design, Setting, and Patients Randomized, double-blind, placebo-controlled, crossover trial (September 2004-March 2007) involving 30 adolescents (aged 11-17 years) who had newly diagnosed, never-treated stage 1 essential hypertension and serum uric acid levels ≥ 6 mg/dL. Participants were treated at the Pediatric Hypertension Clinic at Texas Children's Hospital in Houston. Patients were excluded if they had stage 2 hypertension or known renal, cardiovascular, gastrointestinal tract, hepatic, or endocrine disease.

Intervention Allopurinol, 200 mg twice daily for 4 weeks, and placebo, twice daily for 4 weeks, with a 2-week washout period between treatments. The order of the treatments was randomized.

Main Outcome Measures Change in casual and ambulatory blood pressure.

Results For casual BP, the mean change in systolic BP for allopurinol was -6.9 mm Hg (95% confidence interval [CI], -4.5 to -9.3 mm Hg) vs -2.0 mm Hg (95% CI, 0.3 to -4.3 mm Hg; $P=.009$) for placebo, and the mean change in diastolic BP for allopurinol was -5.1 mm Hg (95% CI, -2.5 to -7.8 mm Hg) vs -2.4 (95% CI, 0.2 to -4.1 ; $P=.05$) for placebo. Mean change in mean 24-hour ambulatory systolic BP for allopurinol was -6.3 mm Hg (95% CI, -3.8 to -8.9 mm Hg) vs 0.8 mm Hg (95% CI, 3.4 to -2.9 mm Hg; $P=.001$) for placebo and mean 24-hour ambulatory diastolic BP for allopurinol was -4.6 mm Hg (-2.4 to -6.8 mm Hg) vs -0.3 mm Hg (95% CI, 2.3 to -2.1 mm Hg; $P=.004$) for placebo. Twenty of the 30 participants achieved normal BP by casual and ambulatory criteria while taking allopurinol vs 1 participant while taking placebo ($P<.001$).

Conclusions In this short-term, crossover study of adolescents with newly diagnosed hypertension, treatment with allopurinol resulted in reduction of BP. The results represent a new potential therapeutic approach, although not a fully developed therapeutic strategy due to potential adverse effects. These preliminary findings require confirmation in larger clinical trials.

Trial Registration clinicaltrials.gov Identifier: NCT00288184

JAMA. 2008;300(8):924-932

www.jama.com

moles per liter, multiply by 59.485), whereas this was observed in only 30% with secondary hypertension (mean, 4.3 mg/dL; $n=40$) and none of the con-

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CRITICAL REVIEWS IN FOOD SCIENCE AND NUTRITION
2020, VOL. 60, NO. 1, 1-10
<https://doi.org/10.1080/10408398.2018.1503155>



REVIEW



Consumption of sugar sweetened beverages and dietary fructose in relation to risk of gout and hyperuricemia: a systematic review and meta-analysis

Soraiya Ebrahimpour-koujan^a, Parvane Saneel^b, Bagher Larijani^c, and Ahmad Esmailzadeh^{a,b,d}

Conclusion: Uric acid directly related to Hypertension.

Chronic Fructose exposure promote METABOLIC SYNDROME

METABOLIC SYNDROME

- **Obesity**
- **Type 2 Diabetes**
- **Lipid problems**
- **Hypertension**
- **Cardio vascular disease (CVD)**

PLEASE REMEMBER



Calorie in which form is important

Obesity = Calorie (not quite true)

25 calories	
 1 cup grape tomatoes	or  2 french fries
100 calories	
 1 cup fresh grapes	or  1 1/2 Tbsp of M&M candies
150 calories	
 6 oz fruited yogurt	or  1 1/2 oz natural cheese
200 calories	
 4 oz pan-seared cod	or  2 oz batter-fried cod
300 calories	
 2 cups minestrone	or  1/16 pepperoni & sausage pizza

Remember...

Fructose $\neq$ Glucose



10 LIVER CLEANSING FOODS



BROCCOLI



GRAPEFRUIT



WALNUTS



LEMONS



BRUSSEL
SPROUTS



AVOCADOS



GARLIC



BEETS



LEAFY
GREENS



GREEN TEA

Food industry: Adulteration of food

1. Fructose is added

- Palatability (especially with low fat food)
- Browning agent (Caramelization)

2. REMOVAL OF FIBER

- Shelf life
- Freezing

3. Transformation of fat in food (Cis-Fat → Trans-fat)

- Shelf life, Hardening agent
- Now removed due to CVD risks





Robert Lustig: How Worried Should We Be About Sugar?

<https://youtu.be/gmC4Rm5cpOI>

Robert H. Lustig is an American pediatric endocrinologist. He is Professor of Pediatrics in the Division of Endocrinology at the University of California, San Francisco, where he specializes in neuroendocrinology and childhood obesity

Take home

1. Bread, alcohol, Fructose metabolism in liver

* Chronic Fructose exposure promote METABOLIC SYNDROME

- Obesity
- Type 2 Diabetes
- Lipid problems
- Hypertension
- Cardio vascular disease (CVD)

2. We intake fructose on a daily basis. **THUS, VERY DIFFICULT TO CONTROL**

3. Obesity $\neq$ Calorie. **CALORIE IS WHICH FORM IS IMPORTANT**

4. Food industry: Adulteration of food

- **Fructose is added**
- **REMOVAL OF FIBER**

5. Good Food = *Fiber rich low sugar food*

In case if you are interested in names

Insulin receptor substrate 1 (IRS-1)

Sterol regulatory element-binding transcription factor 1 (SREBF1)

ATP citrate lyase (ACL),

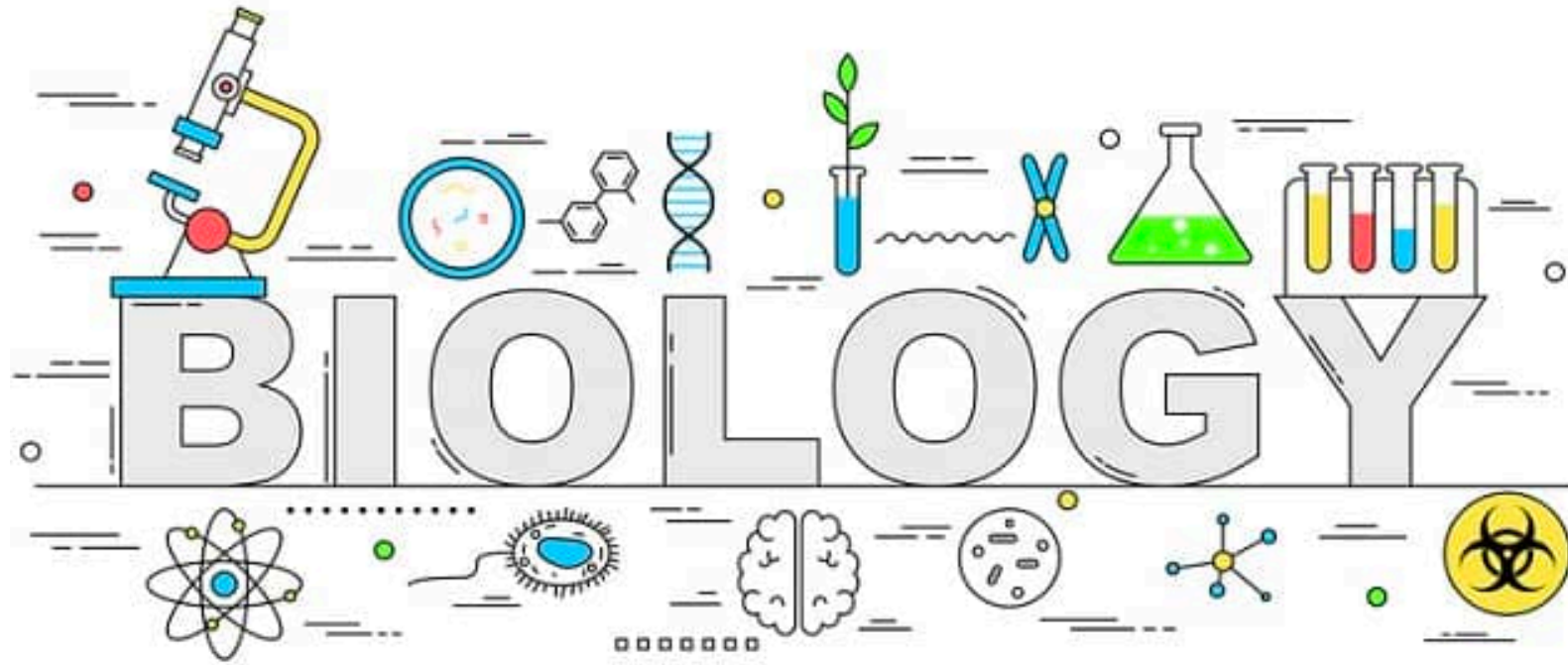
acetyl-CoA carboxylase (ACC)

fatty acid synthase (FAS)

c-Jun N-terminal kinases (JNKs)

Carbohydrate-response element-binding protein (ChREBP)





Next: Proteins